Sentiment Analysis of Rap Music

Leo Xiong, Kaycie Barron, Ryan Unruh, Taewon Min

LIS 501: Introduction to Text Mining

Ming Jiang

Introduction

This study investigates how pretrained sentiment analysis models interpret emotional language in rap lyrics. Sentiment analysis is a widely used technique in natural language processing (NLP) for identifying the emotional tone expressed in text (Liu, 2012). Early work in sentiment analysis primarily focused on domains such as product reviews and opinion mining, where sentiment expression is relatively explicit and literal (Pang & Lee, 2008). More recent approaches incorporate neural language models and contextual embeddings, which have improved systems' ability to capture semantic relationships and contextual meaning in text (Devlin et al., 2019). These advances have enabled models to perform well on standardized benchmarks, often achieving high accuracy in domains where sentiment is clearly signaled through lexical cues.

Despite these improvements, sentiment analysis models are typically trained on datasets derived from domains such as reviews, news articles, or social media posts (Kayed et al., 2020). These domains tend to privilege direct and literal expressions of sentiment, which differ substantially from the artistic and expressive language found in music lyrics. Rap lyrics, in particular, frequently incorporate slang, metaphor, wordplay, and culturally specific linguistic features that challenge surface-level interpretation. As a result, words that appear negative according to traditional sentiment analysis techniques may instead convey positive, ironic, or empowering meanings depending on context. Prior work has also shown that sentiment lexicons vary significantly across domains, meaning that models trained on one domain may systematically misinterpret sentiment in another (Hamilton et al., 2016). This challenge is compounded by evidence that NLP models exhibit bias when processing dialectal language, including African American English (Sap et al., 2019), which plays a central role in rap lyricism.

Accordingly, this project addresses the following research question: To what extent do pretrained sentiment analysis models accurately represent the multi-layered and culturally situated emotional meanings present in rap lyrics, and what systematic limitations emerge in their interpretation? We pay particular attention to cases involving metaphor, cultural language use, and mixed or conflicting emotional signals. We argue that current models fail due to domain mismatch and an underlying assumption that sentiment is singular and lexically grounded, whereas rap lyrics construct meaning through performance, narrative voice, and context-dependent interpretation. This gap is important because it limits the applicability of widely used NLP tools for analyzing culturally rich and expressive forms of language, potentially reinforcing misinterpretations or biases.

Literature Review

A central contribution to this research area is Multi-Emotion Classification for Song Lyrics (Edmonds & Sedoc, 2021), which investigates the challenge of classifying complex emotional signals in song lyrics. The authors demonstrate that ambiguity in lyrics and variability in human annotation perspectives both significantly impact model performance. A key finding is that generalization from out-of-domain data remains a major limitation. Transformer-based models such as BERT perform poorly when trained on social media or dialogue datasets and evaluated on song lyrics, compared to a domain-specific BERT model trained on lyric-specific data. The accuracy of these models fluctuates depending on the composition of the training data.

More broadly, existing sentiment analysis systems are typically optimized for binary or ternary polarity (i.e., positive, negative, neutral), limiting their ability to capture multi-dimensional emotional states such as anticipation, nostalgia, or surprise (Jongeling et al., 2017). Edmonds & Sedoc (2021) therefore frame emotion classification as a multi-label problem, aiming to achieve performance comparable to human inter-annotator agreement. To address the lack of high-quality lyrical datasets, they introduce the Edmonds Dance Dataset, a crowdsourced corpus annotated for eight core emotions based on Plutchik's theory of emotion (Plutchik, 2001). Plutchik's framework conceptualizes emotion as a set of eight primary categories (i.e., joy, sadness, anger, fear, trust, disgust, surprise, and anticipation) that can combine to form more complex emotional states. These emotions are often represented in a "wheel," where proximity reflects similarity and combinations produce nuanced affective experiences. They also reannotated existing multimodal datasets (e.g., Mihalcea & Strapparava, 2012) to enable comparison across annotation schemes. Annotation reliability was evaluated using inter-annotator agreement metrics such as Cohen's Kappa and Krippendorff's Alpha, with low-quality annotations (i.e., those below the 0.25 threshold) filtered out. The authors compare transformer-based models to traditional baselines, including Naive Bayes, Random Forest, and Most Frequent Sense classifiers.

Their results show that state-of-the-art models trained on out-of-domain datasets fail to classify emotions in song lyrics accurately. Notably, BERT models trained on large-scale datasets such as tweets or dialogue perform adequately for high-frequency emotions (e.g., joy, sadness) but are outperformed by simpler models like Naive Bayes for less frequent or more ambiguous emotions (e.g., fear, disgust). Models trained on smaller, in-domain lyrical datasets achieve comparable or superior performance despite significantly less data, a pattern that holds across datasets varying in size, annotation method, and musical genre. This finding highlights the importance of domain alignment over data volume. Overall, Edmonds & Sedoc (2021) provide strong evidence that domain-specific data is preferred for effective emotion classification in song lyrics, and that general-purpose NLP models may struggle with artistic and culturally embedded language.

Similarly, recent work on linguistic complexity in hip-hop lyrics shows that the genre contains dense rhyme schemes, evolving vocabulary, and culturally specific linguistic patterns that reflect broader social and historical changes (e.g., studies of hip-hop corpora using NLP techniques). These findings reinforce the idea that standard NLP models may struggle with this domain due to its linguistic richness and variability. The present work extends this line of research by focusing specifically on sentiment interpretation rather than structural or topical features.

Diachronic analyses of song lyrics further demonstrate that lyrical language has become more repetitive and, in some cases, simpler over time, reflecting a general shift in music production and audience preferences (Interiano et al., 2018). While these studies quantify linguistic trends, they also note that artistic meaning cannot be fully captured through surface-level metrics. The present approach addresses this gap by examining how emotional meaning is interpreted, particularly in cases where surface-level cues are misleading.

In the domain of sentiment modeling, alternative approaches such as sentiment vector space models (Xia et al., 2008) represent lyrics based on emotional features rather than traditional bag-of-words representations. This work identifies challenges such as word ambiguity and limited contextual information, which are especially relevant for lyrical analysis. These approaches still assume that sentiment is lexically grounded, an assumption that is insufficient for rap lyrics, given that meaning in rap lyrics often depends on narrative voice, cultural context, and polarity inversion.

Taken together, prior work demonstrates three key gaps: (1) sentiment models struggle to generalize to lyrical domains (Edmonds & Sedoc, 2021), (2) existing computational approaches focus primarily on structure or lexical features rather than meaning, and (3) artistic language introduces ambiguity that challenges standard NLP assumptions. This project builds on that literature by empirically evaluating pretrained sentiment models on rap lyrics and identifying systematic failure modes, particularly in cases involving mixed sentiment, metaphor, and culturally specific language.

Methods

For our project, we used the *Rap Lyrics for NLP dataset* from Kaggle (Ceebloop, 2020). Our dataset consists of 530 songs from artists including Drake, Kendrick Lamar, Eminem, Nas, Nicki Minaj, and 2Pac. Each artist is represented by roughly 47 to 50 songs (mean = 48). All artists are commercially established English-language rap or hip-hop acts, ranging from 1990s legacy artists like 2Pac through contemporary figures like Dave and Future. The dataset does not include underground, regional, or non-English rap, which limits the scope of any conclusions we can draw. Each entry has four fields: the track name, the artist, the full lyrics scraped from Genius (raw_lyrics), and a cleaned version restricted to verses by the credited main artist (artist_verses).

From this 530-song dataset, we randomly sampled 50 songs across four raters for human annotation. The resulting sample happens to include songs from all eleven artists, but with uneven representation: counts range from three (2Pac, Eminem, Rapsody) to eight (Dave), with a median of four. This annotated subset is our evaluation set. We did not apply a train-test split as we are comparing the outputs of two pretrained models against human judgment, not training a classifier of our own, so the 50 annotated songs function entirely as a held-out test set. A train-test split would only be needed if we were training or fine-tuning a model.

For each song, each annotator assigned a primary valence label from one of four categories (Positive, Negative, Mixed, or Neutral-unclear) and a dominant emotion category from a fixed set (joy/celebration, anger/aggression, sadness/loss, love/affection, resilience/defiance). The annotator also wrote a free-text note describing any interpretive features that affected the label. These features included slang or AAVE usage, irony, narrative voice, and polarity inversion (cases where lexically negative language carries a positive or empowering meaning in context). The notes were not used as input to either model--we kept them for the qualitative error analysis described below, since they let us ask not just whether a model got a song wrong but what kind of language it had trouble with.

We compared two pretrained transformer models from the Hugging Face hub, both built on the RoBERTa architecture (Liu et al., 2019). The first is a general model that is fine-tuned on tweets for three-class sentiment classification (Negative, Neutral, Positive). The second is a DistilRoBERTa model fine-tuned on emotion-labeled corpora to produce one of seven categories: joy, surprise, neutral, sadness, anger, fear, or disgust. We call this the lyric-adjacent or domain-relevant model. It is not trained on rap lyrics. To our knowledge, no rap-specific pretrained sentiment model is publicly available, so this was our closest approximation. Its emotion-focused training objective is closer to the affective content of lyrics than tweet polarity. To compare its outputs with our four-class human labels, we mapped the seven emotions to valence following standard practice in discrete-emotion lexicons (Mohammad & Turney, 2013): joy and surprise mapped to Positive; sadness, anger, fear, and disgust mapped to Negative; neutral mapped to Neutral-unclear.

Both models inherit RoBERTa's 512-token input limit, so we truncated each song's lyrics to the first 512 characters before passing them through the Hugging Face pipeline API for text classification. Each song received one valence prediction from each model. We then relabeled "Neutral" outputs as "Neutral-unclear" so the model labels would match our four-class human scheme. Note that neither model is able to output a "Mixed" prediction at all. We did not try to work around this. One of our questions is whether off-the-shelf models can represent mixed or conflicting emotional signals in the first place, so this gap is part of what we wanted to measure.

We evaluated each model against the human valence labels in three ways. First, using sklearn.metrics.classification_report, we computed per-class precision, recall, and F1-score, along with macro-averaged F1 and overall accuracy. We report macro-F1 because the 50-song sample is imbalanced (Mixed n=17, Positive n=18, Negative n=10, Neutral-unclear n=5), and macro-averaging weights each class equally regardless of size. Second, we generated 4×4 confusion matrices for each model to see which human-labeled categories were collapsing into which predicted ones. Third, for the 34 songs that both models got wrong at the exact-label level, we performed a qualitative error analysis: each error was tagged against the linguistic features recorded in the annotation notes (slang/AAVE, polarity inversion, irony/sarcasm, metaphorical/performative language, and narrative voice).

Beyond the classification step, we also wanted to see how each model represented songs internally. For each song, we ran the lyrics through the model's tokenizer (max length 512, truncated) and took a mean-pooled vector across the final hidden layer's token embeddings, which gave us a 768-dimensional vector per song per model. We then used t-SNE (van der Maaten & Hinton, 2008) with perplexity = 15 and random_state = 42 to project these vectors down to two dimensions. Each song was colored by its human-assigned valence label. The idea is that if a model represents sentiment in a way that aligns with human judgment, songs with the same human label should land near each other on the plot. If they don't, the points will look randomly mixed.

Results

Evaluation was conducted on a human-annotated subset of 50 songs using a four-class valence scheme (Positive, Negative, Mixed, Neutral-unclear). All model outputs, confusion matrices, predictions, and error analysis files are available at: https://github.com/unroooo/LIS-501/tree/main/kaycieresults
Model performance is summarized below:

Table 1. *General model classification results, four-class valence scheme*

Class	**Precision**	**Recall**	**F1-score**	**Support**
Negative	0.33	0.60	0.43	10
Neutral-unclear	0.12	0.60	0.19	5
Mixed	0.00	0.00	0.00	17
Positive	0.67	0.22	0.33	18
Accuracy			0.26	50

Macro avg	0.28	0.36	0.24	50
Weighted avg	0.32	0.26	0.23	50

The general model shows limited alignment with human judgment across all classes. Most notably, it completely fails to identify Mixed sentiment (F1 = 0.00), despite this being one of the largest categories in the dataset. This indicates that the model is unable to represent co-occurring or conflicting emotional signals and instead collapses them into simpler categories, such as Neutral or Negative.

The model also demonstrates class imbalance. While precision for Positive is relatively high (0.67), recall is low (0.22), indicating that many positive songs are not being correctly identified. At the same time, recall for Negative (0.60) suggests the model is more sensitive to negative lexical cues. Overall, the low accuracy (0.26) and macro F1 (0.24) reflect difficulty generalizing from social media-style sentiment to lyrical data.

Table 2. *Lyric-adjacent model classification results, four-class valence scheme*

Class	**Precision**	**Recall**	**F1-score**	**Support**
Negative	0.29	0.80	0.42	10
Neutral-unclear	0.13	0.40	0.20	5
Mixed	0.00	0.00	0.00	17
Positive	0.29	0.11	0.16	18
Accuracy			0.24	50
Macro avg	0.18	0.33	0.20	50
Weighted avg	0.17	0.24	0.16	50

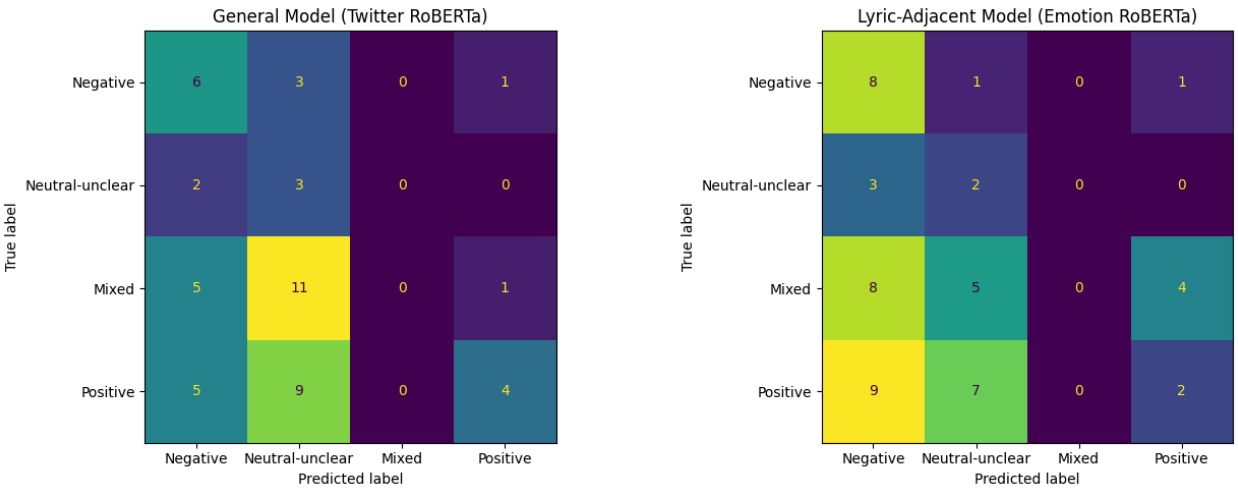
The lyric-adjacent model does not substantially improve performance despite being trained on emotion-focused data. Like the general model, it completely fails to capture Mixed sentiment (F1 = 0.00), reinforcing the limitation that pretrained models assume a single dominant emotional label.

This model shows a strong bias toward Negative classification, with very high recall (0.80) but low precision (0.29). This indicates that many songs are being over-classified as negative, likely due to the presence of aggressive or explicit language. Positive performance is particularly weak (F1 = 0.16),

suggesting difficulty recognizing celebratory or success-oriented themes when they are expressed through stylistic or culturally specific language.

Overall, the lower macro F1 (0.20) suggests that adding emotional granularity does not resolve the underlying mismatch between the model's assumptions and lyrical meaning.

Figure 1. *Confusion matrices for both models, four-class valence scheme*



For the General Model, correct predictions are concentrated in Negative (6/10) and Neutral-unclear (3/5), but Mixed is never predicted. Most Mixed songs are reassigned to Neutral-unclear (11 cases) or Negative (5 cases), while Positive songs are frequently misclassified as Neutral-unclear (9) or Negative (5).

For the Lyric-Adjacent Model, Negative predictions dominate (8/10 correct), but again, Mixed is never predicted. Mixed songs are distributed across Negative (8), Neutral-unclear (5), and Positive (4), while Positive songs are primarily misclassified as Negative (9) or Neutral-unclear (7).

Across both models, the dominant pattern is a systematic collapse of the Mixed category and a redistribution of both Mixed and Positive songs into Negative or Neutral-unclear. The lyric-adjacent model exhibits a stronger tendency toward Negative classification, whereas the general model more frequently defaults to Neutral-unclear. In both cases, the models fail to preserve the multi-dimensional structure of human-labeled sentiment.

Table 3. *Overall error counts for both models*

Metric	**Value**
Total songs	50

General model errors (4-class exact match)	37
Lyric-adjacent model errors (4-class exact match)	38
Songs misclassified by both models	34
Cases where the general model was correct, and lyric-adjacent was not	4
Cases where the lyric-adjacent model was correct, and the general model was not	3

Both models perform similarly, with high error rates and minimal differences in exact-match accuracy. The fact that both models are incorrect on 34 out of 50 songs indicates that errors are not model-specific but systematic. This suggests a shared limitation in how sentiment is represented, rather than a difference in model quality.

Table 4. *Sample misclassifications from both models*

Song	**Human**	**cardiffnlp/twitter-roberta-base-sentiment**	**j-hartmann/emotion-english-distilroberta-base**	**Interpretation**
Codeine Crazy – Future	Mixed	Neutral	Neutral	Addiction framed positively → polarity inversion + layered meaning
Shake That – Eminem	Positive	Negative	Negative	Explicit language dominates → lexical bias
Location – Dave	Positive	Neutral	Neutral	Success expressed through slang → missed cultural cues
Let Me Calm Down – Nicki Minaj	Mixed	Neutral	Negative	Conflicting emotions → multi-emotion collapse

Plain Jane Remix – Nicki Minaj	Positive	Neutral	Negative	Braggadocio interpreted as negative → performance voice

Misclassifications consistently occur in cases where meaning is not directly tied to surface-level sentiment. Songs involving mixed emotional tone, narrative framing, or culturally specific language are reduced to simpler categories. In particular, polarity inversion is a recurring source of error.

Table 5. *Distribution of error types across the songs misclassified by both models*

Error type	**Count**
Other	9
Metaphorical/performative language	8
Polarity inversion	7
Slang/AAVE	6
Narrative voice	2
Irony/sarcasm	2

The distribution of error types highlights that failures are systematic rather than random. The most common error categories—metaphor, polarity inversion, and slang/AAVE—require contextual and cultural interpretation beyond lexical sentiment cues. This suggests that both models rely heavily on surface features and are not well-equipped to interpret meaning constructed through performance, identity, and/or narrative voice.

Across both evaluation settings, the lyric-adjacent emotion model does not outperform the general sentiment model, despite being trained on affect-rich data. One explanation lies in differences in task formulation and training data. The general model (Twitter RoBERTa) is explicitly fine-tuned for sentiment polarity classification (negative, neutral, positive) on a very large corpus of social media text (tens of millions of tweets). In contrast, the emotion model is trained to predict seven discrete emotion categories (e.g., joy, anger, sadness) across multiple datasets and is based on a distilled architecture (DistilRoBERTa), which reduces model size and complexity.

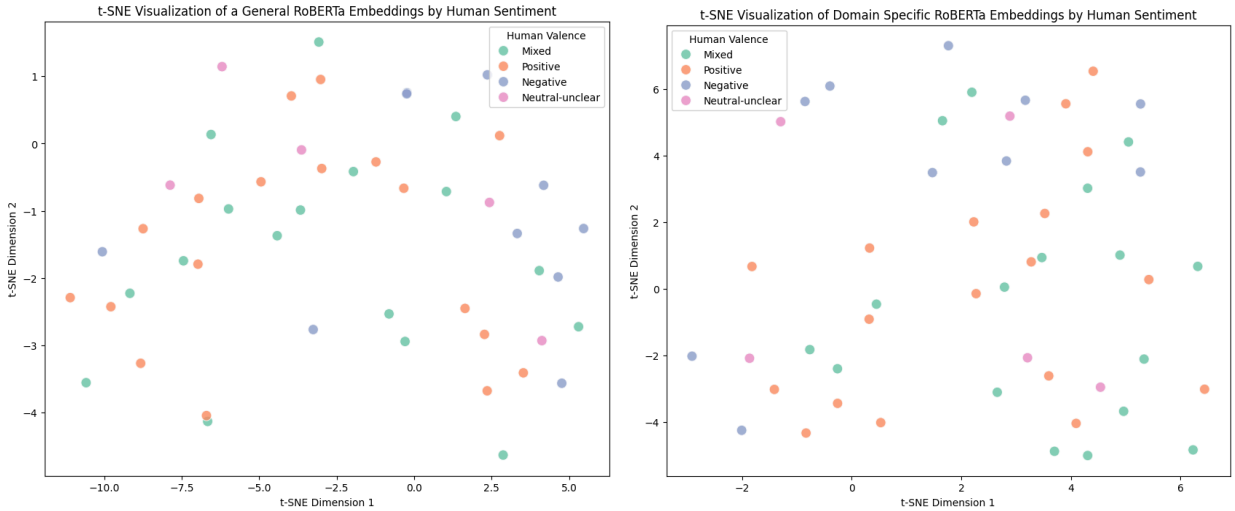
This distinction is consequential. The emotion model must first map text into fine-grained emotional categories and is then indirectly collapsed into valence, introducing an additional layer of

abstraction and potential information loss. Meanwhile, the general model is optimized directly for polarity, aligning more closely with the evaluation task.

More importantly, both models are trained on domains where sentiment is typically lexically grounded and singular, whereas rap lyrics often encode meaning through performance, narrative stance, and polarity inversion. The emotion model's sensitivity to emotionally salient words (e.g., aggression, fear) appears to increase its tendency to overpredict Negative, while the general model more often defaults to Neutral when signals are ambiguous.

Taken together, these results suggest that model specialization alone does not guarantee better performance. Instead, alignment between training objective, label space, and domain-specific meaning-making practices is more critical. In this case, neither model adequately captures the multi-layered, context-dependent nature of sentiment in rap lyrics.

Figure 2. *t-SNE projections of song embeddings for both models, colored by human-assigned valence*



We also looked at how each model represents songs internally, before the final classification step. For each song, we pulled the model's 768-dimensional embedding and used t-SNE to project it down to two dimensions for plotting. Each point is one song, colored by its human-assigned valence label.

In both plots, songs with the same human label do not group; songs (green) are spread across the full width of each plot, often sitting right next to Positive (orange) and Negative (blue) points. The five Neutral-unclear songs (pink) are scattered throughout. No part of either plot is mostly one color. The two plots also look about the same as each other. The lyric-adjacent model does not produce tighter or cleaner groupings than the general model, even though it was trained on emotion data.

Discussion

Our results strongly align with the foundational limitations of out-of-domain generalization identified by Edmonds and Sedoc (2021). While their work highlighted that models trained on social media or dialogue datasets fail to accurately classify emotions in song lyrics, our error analysis of 34 mutually misclassified songs, as shown in Table 3, extends this by demonstrating why this domain mismatch occurs in rap music. We found that standard NLP models fundamentally lack the cultural and contextual frameworks necessary to process multi-layered artistic meaning. Because both of our evaluated models relied heavily on surface-level lexical cues, they exhibited systematic failures when processing metaphorical language, slang, and African American Vernacular English (AAVE), as seen in Table 5. Furthermore, Table 4 shows that the models were consistently confounded by polarity inversion, proving that model specialization alone cannot overcome a lack of domain-specific cultural context.

Additionally, our models' complete failure to identify "Mixed" sentiment across all evaluations in Tables 1 and 2 reinforces Edmonds and Sedoc's critique that most current systems are artificially limited by their assumption of a single dominant emotional label. Songs frequently exhibit a coexistence of multiple, sometimes conflicting, emotions. By forcing a singular label, our models systematically collapsed these complex lyrical signals into inaccurate, simplistic categories like Neutral-unclear or Negative, as evidenced by our confusion matrices in Figure 1 and the scattered, overlapping nature of our song embeddings in Figure 2.

Finally, our evaluation supports the existence of a steep hierarchy of difficulty in emotion classification. Just as Edmonds and Sedoc observed that F1-scores for darker, nuanced emotions like anger, disgust, and fear remain consistently low across various state-of-the-art models, our models exhibited the same struggle, frequently overpredicting or misclassifying these states based on explicit language rather than actual sentiment, which can be seen in the results in Table 4. This ambiguity is most pronounced with the emotion of surprise, which Edmonds and Sedoc note is exceptionally difficult to model due to a lack of human consensus. They found that surprise yields a remarkably low inter-annotator agreement, producing a Krippendorff’s Alpha of just 0.532 (Edmonds & Sedoc, 2021, Table 3). As our findings also suggest, when human annotators cannot establish a reliable ground truth for complex lyrical expressions, general-purpose predictive models will inevitably fail to interpret them correctly

Our study presents several methodological and data-related limitations. First, our evaluation relied on a relatively small, human-annotated subset of only 50 randomly sampled songs. This sample also exhibited class imbalance, heavily skewing toward "Mixed" and "Positive" labels while underrepresenting "Neutral-unclear". Furthermore, because both evaluated models inherit the RoBERTa architecture's 512-token input limit, all lyrics had to be truncated. This truncation may have excluded vital narrative shifts or emotional context found later in the tracks. Methodologically, the absence of a publicly

available, rap-specific pretrained sentiment model forced our reliance on off-the-shelf approximations. This introduced significant structural constraints; most critically, neither model was capable of outputting a "Mixed" prediction. Additionally, the lyric-adjacent emotion model required us to map its seven discrete emotional categories down to our four-class valence scheme, a process that inherently introduces an extra layer of abstraction and potential information loss.

Addressing the systemic failures of sentiment models in artistic domains remains a significant open challenge. Current computational approaches are too heavily lexically grounded and struggle to navigate the ambiguity, narrative voice, and cultural nuances intrinsic to rap music. To overcome these hurdles and improve the applicability of NLP tools in culturally rich domains, future solutions should consider the following guidelines:

1. Models must be fine-tuned on genre-specific lyrical corpora rather than general social media text. Alignment between the training data and the domain's unique meaning-making practices is far more critical than merely using affect-rich data.
2. To prevent the artificial collapse of complex emotions, sentiment classification in music should be framed as a multi-label problem. This allows models to identify and represent the coexistence of multiple, sometimes conflicting, emotional states within a single song.
3. Future architectures must develop mechanisms to detect polarity inversion and contextual irony, ensuring that narrative stance and performance are weighted as heavily as surface-level vocabulary.

Conclusion

In short, the domain shapes the model's behavior by defining the linguistic patterns it learns. For example, in Twitter feeds or product reviews, sentiment is often tied to isolated words; "bad" or "hate" typically signal a negative sentiment. However, rap lyrics challenge this logic by using the same vocabulary to convey confidence or resilience through context. The models are still picking up on the surface-level cues, but they're missing how the meaning is being constructed across the verse or the whole song.

Another big takeaway is how the models handled mixed sentiment. In our dataset of 50 human-annotated songs, 17 were labelled as mixed, the largest category that was in our annotated dataset, but neither model predicted mixed at all. Those songs mostly got pushed to negative or neutral/unclear, which simplifies some things that are clearly more layered than human annotations. We also saw consistent patterns in the types of misclassifications, slang, metaphors, and polarity inversions. For example, let the songs that describe something that is difficult, but framed more confidently or positively, be interpreted as a negative. There were also cases where the exaggeration of the lyrics was taken literally.

We expected the emotion-based model to help with some of this since it has more fine-tuned categories, but the results are still similar between the two models, the domain-specific and the general models. This suggests that the issue goes beyond the label and how it is tied to the meaning overall. Despite different labels, these models seem to operate on the same faulty assumptions about emotion. Their tendency to fail on the same tracks points to a shared blind spot: an inability to look past surface-level keywords to understand the actual intent behind the lyrics.

The final takeaway involves evaluation. Metrics such as accuracy and F1 scores offer an incomplete picture; comparing model predictions to human interpretation adds much-needed depth. While human annotation is the gold standard, it has its limits, as individual perspectives on emotion can vary. However, the annotation notes helped explain why certain predictions didn't align, providing a level of detail that metrics alone often miss.

Given additional time and computational resources, future iterations of this research would transition from evaluating pretrained models to developing a custom, domain-specific classifier. This approach would necessitate compiling a significantly larger training corpus composed exclusively of rap lyrics, ensuring the model effectively learns the genre's unique linguistic framework. Furthermore, the classification pipeline would be reconfigured to support multi-label predictions. Such an architectural adjustment would enable the accurate representation of mixed or concurrent emotional states, addressing the artificial constraints imposed by single-label classification. Finally, future work would investigate the integration of explicit contextual features, such as domain-adapted lexicons or narrative stance embeddings. These additions would be designed to enhance the model's capacity to detect polarity inversion and irony, ultimately mitigating the erroneous literal interpretation of performative language.

References

Ceebloop. (2020). Rap lyrics for NLP (Version 1). Kaggle. https://www.kaggle.com/datasets/ceebloop/rap-lyrics-for-nlp

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (pp. 4171–4186). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Edmonds, D., & Sedoc, J. (2021). Multi-emotion classification for song lyrics. In O. De Clercq, A. Balahur, J. Sedoc, V. Barriere, S. Tafreshi, S. Buechel, & V. Hoste (Eds.), Proceedings of the Eleventh

Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis (pp. 221–235). Association for Computational Linguistics. https://aclanthology.org/2021.wassa-1.24

Guellil, I., Azouaou, F., Benali, F., & Ala-Eddine, H. (2021). A dataset for detecting real-world empathy and distress. In O. De Clercq, A. Balahur, J. Sedoc, V. Barriere, S. Tafreshi, S. Buechel, & V. Hoste (Eds.), Proceedings of the Eleventh Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis (pp. 236–249). Association for Computational Linguistics. https://aclanthology.org/2021.wassa-1.24

Hamilton, W. L., Clark, K., Leskovec, J., & Jurafsky, D. (2016). Inducing domain-specific sentiment lexicons from unlabeled corpora. In Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (pp. 595–605). Association for Computational Linguistics. https://doi.org/10.18653/v1/D16-1057

Interiano, M., Kazemi, K., Wang, L., Yang, J., Yu, Z., & Komarova, N. L. (2018). Musical trends and predictability of success in contemporary songs. Royal Society Open Science, 5(5), Article 171274. https://doi.org/10.1098/rsos.171274

Jongeling, R., Sarkar, P., Datta, S., & Serebrenik, A. (2017). On negative results when using sentiment analysis tools for software engineering research. Empirical Software Engineering, 22(5), 2543–2584. https://doi.org/10.1007/s10664-016-9498-y

Kayed, M., Díaz-Redondo, R. P., & Mabrouk, A. (2020). Deep learning-based sentiment classification: A comparative survey. IEEE Access, 8, 85616–85638. https://doi.org/10.1109/ACCESS.2020.2992013

Liu, B. (2012). Sentiment analysis and opinion mining. Synthesis Lectures on Human Language Technologies. https://doi.org/10.2200/S00416ED1V01Y201204HLT016

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. arXiv. https://doi.org/10.48550/arXiv.1907.11692

Mihalcea, R., & Strapparava, C. (2012). Lyrics, music, and emotions. In Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning (pp. 590–599). Association for Computational Linguistics. https://aclanthology.org/D12-1054

Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. Computational Intelligence, 29(3), 436–465. https://doi.org/10.1111/j.1467-8640.2012.00460.x

Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. Foundations and Trends in Information Retrieval, 2(1–2), 1–135. https://doi.org/10.1561/1500000011

Plutchik, R. (2001). The nature of emotions. American Scientist, 89(4), 344–350. https://doi.org/10.1511/2001.4.344

Sap, M., Card, D., Gabriel, S., Choi, Y., & Smith, N. A. (2019). The risk of racial bias in hate speech detection. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (pp. 1668–1678). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1163

van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. Journal of Machine Learning Research, 9, 2579–2605. http://jmlr.org/papers/v9/vandermaaten08a.html

Xia, Y., Wang, L., Wong, K.-F., & Xu, M. (2008). Lyric-based song sentiment classification with sentiment vector space model. In Proceedings of ACL-08: HLT, Short Papers (pp. 133–136). Association for Computational Linguistics. https://aclanthology.org/P08-2034

Acknowledgements

Group 1 was formed with the initial triage of Leo Xiong, hoping to have peers enter their names and contact information. Shortly after, Kaycie Barron, Ryan Unruh, and Taewon Min joined to officially form group 1. We chose between analyzing fake news and sentiment analysis of music. Originally, we were going to compare two genres, R&B and R&B Rap, but we agreed that was too much and stuck to rap.

Deliverable #1 - Leo Xiong came up with general ideas and a discussion on how we wanted to present our topic, and what topic we wanted. The group added ideas and formatted the paper and presentation to make it look cleaner. After our proposal, Kaycie Barron updated our deliverable, along with writing out the feedback that Ming gave us in order to improve our research.

Deliverable #2 - For the 50 Article annotations, each member had to do their own research to find articles that fit our proposed topic, and come up with a summary that fit each article. Approximately, each member found 12-13 articles.

Out of our 50 references, Kaycie suggested one article, which we all agreed on and Leo Xiong created deliverable #2 based on the suggested paper. Ryan Unruh used this information and created our presentation based on the information that Kaycie and Leo Xiong took from the article. After the creation of the presentation, the group reviewed it and confirmed it was satisfactory.

Deliverable #3 - Kaycie Barron did the analyses in Python except for t-SNE, wrote the introduction, and wrote the literature review section. Leo Xiong also did a version of the analyses in Python, and Ryan Unruh did the code to generate the t-SNE. Leo Xiong created the presentation template, and group members chose their sections and filled them out. We each recorded ourselves discussing different sections, and Leo Xiong edited the videos together.

Final Code - Our codebase is on GitHub, started by Ryan Unruh, and group members collaborated on it. Our codebase contains the original dataset that we found via Kaggle, group members' individual work, and our human-annotated dataset.

For our human-annotated dataset, Leo Xiong annotated 27 songs, Ryan Unruh annotated 14 songs, Kaycie Baron annotated 13 songs, and Taewon Min annotated 0 songs. Each song was randomly selected by Ryan Unruh. Ryan Unruh also created the template for annotating the music. After annotating each song, Leo Xiong added them to one sheet and created a .csv file out of them to create our test dataset, where results were found.